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PROJECT PROPOSAL REPORT (REVISED)

Improving Classroom HVAC Energy Efficiency through Personalized Thermal Comfort Data Analytics: A Field Study and Simulation Assessment

Project ID: FX02

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1. Background

Buildings account for between 20% and 40% of final energy use in developed economies, and heating, ventilation and air-conditioning (HVAC) is the largest single end use in non-domestic buildings [1]. In cooling-dominated cities such as Hong Kong, air-conditioning in commercial and institutional buildings is a particularly important load. A substantial share of this energy is spent conditioning spaces in which occupants are still not satisfied, and closing this gap between energy input and occupant satisfaction is a central aim of smart, occupant-aware HVAC control [2]. The problem is especially relevant to educational buildings: classrooms have high and variable occupant densities for long periods, and classroom thermal conditions have been shown to affect students' performance on schoolwork [3].

Smart HVAC controllers promise to replace static setpoints, usually derived from population-level comfort models, with setpoints informed by feedback from the actual occupants. The underlying logic is simple: a controller that knows how the people in a room respond can deliver comfort with less energy than one that conditions the space for an assumed average occupant. The statistical foundation of this personalisation is, however, weaker than it appears. Thermal sensation is usually recorded on the ASHRAE 7-point scale (cold, cool, slightly cool, neutral, slightly warm, warm, hot), which is an ordinal scale, yet the dominant analytical practice, including recent machine-learning studies such as that of Song et al. [4], treats the votes either as continuous numbers or as unordered classes. Favero et al. [5] showed that analysing such votes with linear regression can change substantive conclusions, distort regression coefficients, and conceal the fact that the distances between adjacent scale categories are not equal. If population-level comfort models are statistically mis-specified, personalised controllers built on them inherit the problem, and the size of the resulting error in HVAC energy and comfort performance is currently unknown.

This project links statistically appropriate, data-driven comfort modelling to HVAC energy performance, using a university classroom as the field site. It has two coordinated parts. In Part 1, approximately twenty students who use the same classroom will cast repeated thermal sensation votes over an eight-week campaign, paired with environmental measurements from three sensor stations distributed across the room. Three comfort models will be fitted to this dataset: a linear (ordinal-as-metric) model representing common practice, a population-level ordinal model, and a personalised hierarchical ordinal model with occupant-specific response parameters. In Part 2, the fitted models will parameterise HVAC controllers in an EnergyPlus model of the classroom, and the annual energy use and comfort performance of each controller will be compared. Together, the two parts translate a statistical modelling choice into quantities that designers of smart HVAC systems for educational buildings can act on.

2. Literature Review

2.1 Conventional Thermal Comfort Models and Their Limitations

The most widely used thermal comfort model is the Predicted Mean Vote (PMV) developed by Fanger [6]. PMV estimates the mean thermal sensation of a large group of people from four environmental variables (air temperature, mean radiant temperature, air velocity and relative humidity) and two personal variables (clothing insulation and metabolic rate). Together with the Predicted Percentage Dissatisfied (PPD), it underpins ASHRAE Standard 55 [7] and ISO

7730 [8], as well as most HVAC setpoint logic used in practice. PMV is, however, a steady-state model derived largely from climate-chamber experiments, and field studies have shown that it can substantially misestimate occupants' sensation in real buildings, particularly in naturally ventilated spaces and in transitional seasons [9]. Adaptive comfort models, such as that of de Dear and Brager [10], partly address this by allowing comfort temperatures to follow outdoor conditions, but they remain population-level models that treat occupants as interchangeable.

2.2 Machine Learning Approaches to Thermal Comfort

In response to these limitations, a growing literature applies machine learning (ML) to predict thermal comfort from occupant feedback and environmental sensing. Song et al. [4] provide a representative example. In a field study of 1,162 respondents in a Chinese university library, they collected more than sixty features describing the thermal environment, physiological characteristics, psychological state and well-being, reduced them by principal component analysis, and compared six classifiers: logistic regression, linear discriminant analysis, k-nearest neighbours, classification and regression trees, Gaussian naive Bayes and support vector machines. Prediction accuracy was low for the full seven-category problem (about 39% at best) and considerably higher once the scale was collapsed to three or two categories (up to about 73%). The study illustrates both the potential of ML for occupant-aware HVAC and a recurring weakness of the field.

The weakness is that most ML-based comfort studies, including [4], evaluate models with nominal metrics such as classification accuracy. These metrics penalise all misclassifications equally: predicting "warm" when the true vote is "hot" counts as the same error as predicting "cold". For an ordinal outcome such as thermal sensation, in which the order of the categories carries meaning, this discards information. Lai and Chen [11] showed directly that ordinal and multinomial regression models predicted individual thermal sensation votes considerably better than linear regression applied to the same data.

2.3 Ordinal Data, Ordinal Methods and the Recent Methodological Critique

Drawing on the statistical literature on ordinal regression [12], [13], Liddell and Kruschke [14] showed that analysing ordinal responses with metric methods such as t-tests, analysis of variance and linear regression can produce four kinds of error: inflated false-alarm rates, reduced correct-detection rates, distorted effect-size estimates and even reversed effects. Favero et al. [5] brought this critique to thermal comfort research. Using a single field dataset of 6,048 votes collected in Indian offices [15], extracted from the ASHRAE Global Thermal Comfort Database II [16], they fitted a Bayesian cumulative probit model and a linear regression model with gender and air temperature as predictors. The two models led to different conclusions about the gender difference; the linear model distorted the air-temperature coefficient; and the ordinal model showed that the thresholds between adjacent categories of the ASHRAE scale are not equidistant. Schweiker et al. [17] reached a compatible conclusion from a large multi-country questionnaire study, finding that groups of respondents differ in how far apart they perceive the scale anchors to be. Favero et al. conclude that ordinal data should be analysed with ordinal models, and note that such models can allow thresholds to depend on covariates — an extension they did not pursue.

2.4 Personalised Thermal Comfort and Smart HVAC Control

In parallel, a separate literature has developed personal comfort models and their use in HVAC control. Kim et al. [18] proposed personal comfort models as a new paradigm: models trained on an individual's own feedback that predict that individual's comfort response, rather than a population average. Approaches range from Bayesian-network models of individual occupants [19] to occupancy- and behaviour-aware model predictive control [2]. Wang et al. [20] reviewed the evidence for substantial individual differences in thermal comfort associated with factors such as gender, age and physiological condition. Where repeated votes from the same people have been analysed with multilevel models, the choice between linear and ordinal mixed-effects regression has again been shown to change conclusions [21]. Personal comfort studies commonly judge the benefit of personalisation against a population-level baseline. If that baseline is built with ordinal-as-metric methods, as is often the case, the estimated benefit of personalisation carries a bias of unknown size and direction.

2.5 Research Gap

Three gaps emerge from this review. First, the non-equidistance of ASHRAE scale thresholds has been demonstrated at population and group level [5], [17], but, to the author's knowledge, occupant-specific threshold structure has not been estimated from repeated field votes using partially pooled ordinal models — which is precisely what a personalised controller needs. Second, the personal comfort literature has not established how many votes per occupant are required before a personalised model predicts better than a population model, a question of direct practical importance for smart-thermostat design. Third, the choice between ordinal and ordinal-as-metric modelling has not, to the author's knowledge, been propagated through a building energy simulation to quantify its effect on annual HVAC energy use and comfort; classrooms, with their distinctive occupancy and energy profiles, have received particularly little attention in this context. The present project addresses all three gaps.

3. Objectives

The aim of this project is to quantify, through a combined classroom field study and building simulation, the extent to which personalised ordinal comfort models improve thermal comfort prediction and HVAC energy performance relative to conventional population-level approaches. The specific, measurable objectives are:

O1. To collect, under university ethics approval, a repeated-measures dataset from approximately twenty students in a single regular teaching classroom over an eight-week campaign, comprising thermal sensation votes paired with synchronised environmental measurements (air temperature, relative humidity and CO_2 concentration at three stations; globe temperature and air velocity at the central station), with a target of at least 15 valid votes per participant.

O2. To fit and compare three thermal comfort models on this dataset — an ordinal-as-metric linear model (Model A), a population-level cumulative probit model (Model B), and a hierarchical cumulative probit model with partially pooled occupant-specific location, scale and temperature-sensitivity parameters (Model C) — evaluated by ordinal predictive metrics on held-out votes and, between the two ordinal models, by the Leave-One-Out Information Criterion (LOOIC).

O3. To determine the number of votes per occupant at which Model C's predictive performance on each occupant's later votes, measured by Quadratic Weighted Kappa, ordinal Mean Absolute Error and log score, exceeds that of Model B.

O4. To build an EnergyPlus model of a single-zone classroom representative of the field-study room under Hong Kong climate, and to implement three controllers through the EnergyPlus Python API: a fixed-setpoint PMV-based controller (Controller 1), a population-adaptive controller based on Model B (Controller 2) and a personalised-adaptive controller based on Model C (Controller 3).

O5. To simulate the three controllers over a full year and report annual HVAC energy use (kWh, by end use) together with occupied-hour dissatisfaction metrics evaluated under three comfort criteria (PMV/PPD, Model B and Model C), and thereby quantify the energy differences attributable to each modelling choice at comparable comfort.

Each objective yields an auditable output: an ethics-approved dataset, three fitted models with diagnostics, a votes-per-occupant break-even curve, a validated EnergyPlus model, and an annual performance comparison. No objective is phrased as learning, understanding or surveying, in line with the BSE3716 proposal guidelines.

4. Methodology

The methodology is organised in two parts. Part 1 covers the classroom field study and statistical modelling (Objectives 1–3); Part 2 covers the EnergyPlus simulation and controller evaluation (Objectives 4–5). The project runs from October 2026 to May 2027 (Section 5). To reduce schedule risk, the statistical pipeline and the EnergyPlus base model will be built and tested on simulated data before the field data arrive, so that the period after the campaign is spent on analysis rather than on tool development.

4.1 Part 1 – Classroom Field Study and Statistical Modelling

Ethics and consent. Because the study involves human participants, it will be submitted for review through the University's Human Subjects Ethics Application Review System (HSEARS) [22] before any data are collected, including pilot data. The application will include a participant information sheet, an informed consent form, the questionnaire and a data management plan. Because participants are the author's classmates, particular care will be taken to avoid any perception of pressure to take part: participation is voluntary and has no bearing on any academic assessment; the course instructor will not be told who participates; and participants may withdraw at any time without giving a reason, with their data removed if they withdraw before analysis begins. Responses will be identified only by a participant-generated code, and the data will be stored on a secure university drive accessible only to the author and the supervisor. Any additional departmental requirements will also be followed.

Field site and participants. Approximately twenty students from a final-year class that meets in a single regular teaching classroom during Semester 2 of 2026/27 will be invited to participate. The room's geometry, orientation, window area, shading and HVAC equipment will be recorded in an initial site survey, which also supplies the inputs for the EnergyPlus model in Part 2.

Data-collection window and sample size. The main campaign will run for eight teaching weeks, from mid-January to mid-March 2027. Votes will be collected at every class session, twice per session (near the start and roughly halfway through), to capture within-session thermal drift. For a class that meets twice a week this gives up to 32 votes per participant; assuming a completion rate of about 70%, the expected yield is roughly 20 votes per participant and about 400 votes in total. A minimum of 15 valid votes per participant is needed for the break-even analysis described below; if the class meets only once a week, the campaign will be extended or a second class recruited. The January–March window is also expected to cover a wider range of indoor conditions than an autumn campaign, from ventilation-only operation in cooler weeks to cooling operation as the weather warms. A narrow indoor temperature range would make the temperature sensitivity hard to estimate, so the achievable range will be checked during the sensor dry run in December–January.

Sensor instrumentation and placement. Three sensor stations will be deployed: one near the centre of the occupied zone (the reference station), one near the window wall (to capture solar and envelope effects) and one near the internal wall or door (the opposite end of the spatial gradient). Each station will log air temperature, relative humidity and CO_2 concentration at one-minute intervals. Globe temperature and air velocity will be measured at the reference station and used to compute mean radiant temperature and operative temperature following ISO 7726 [23], and to calculate PMV. Before deployment, all loggers will be co-located for at least 24 hours to check agreement, and their specifications will be compared with the accuracy requirements of ISO 7726. Each participant's seat position will be recorded, and each vote will be paired with the nearest station's reading at the vote timestamp. Window and door states and the HVAC operating mode will be noted at each session.

Questionnaire design. The questionnaire, designed by the author for this study, will be delivered through a mobile-friendly web form and take less than 60 seconds to complete. The core item is the ASHRAE 7-point thermal sensation vote; a three-point thermal preference item (want warmer / no change / want cooler) is included as a secondary outcome. Further items record clothing (a garment checklist converted to clo values), activity in the preceding 15 minutes as a proxy for metabolic rate, and seat position. Age band and gender are collected once at enrolment. The questionnaire will be piloted with two or three classmates under the approved ethics protocol and refined before the main campaign.

Statistical models. Votes are coded from 1 (cold) to 7 (hot) for the ordinal models and from −3 to +3 for Model A. The predictor vector x_{ij} for vote i of occupant j contains standardised operative temperature, clothing insulation, activity level and relative humidity. Three models will be fitted:

- **Model A (ordinal-as-metric baseline):** a linear regression of the numeric vote on the predictors, replicating common practice and serving as the reference for mis-specification effects (Eq. 1).
- **Model B (population ordinal):** a cumulative probit model with thresholds and slopes shared by all occupants, following Favero et al. [5] (Eq. 2).
- **Model C (personalised ordinal):** a hierarchical cumulative probit model in which each occupant has a threshold shift u_j, a response-scale parameter s_j and a temperature-sensitivity deviation v_j, all partially pooled towards the population values (Eq. 3).

$$y_{ij} = \alpha + x_{ij}^T\beta + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim N(0, \sigma^2) \tag{1}$$

$$Pr(Y_{ij} \leq k) = \Phi(\tau_k - x_{ij}^T\beta), \quad k = 1, \ldots, 6 \tag{2}$$

$$Pr(Y_{ij} \leq k) = \Phi(\, [\tau_k - x_{ij}^T\beta - u_j - v_j T_{op,ij}] \,/\, s_j\,), \quad s_j = exp(-d_j) \tag{3}$$

Here Φ is the standard normal cumulative distribution function and $\tau_1 < \ldots < \tau_6$ are the shared thresholds. In Model C, (u_j, v_j) follow a bivariate normal distribution with zero mean and $d_j \sim N(0, \sigma_d^2)$. Occupant *j*'s effective thresholds are therefore $\tau_k - u_j$, rescaled by s_j, so the model captures individual differences in where and how widely each person uses the scale without estimating six free thresholds per person. Fully separate thresholds would not be identifiable with around twenty votes per occupant, particularly because few occupants are expected to use the extreme categories; partial pooling shrinks the estimates for occupants with few votes towards the population values.

Following Fanger's convention, an occupant is counted as dissatisfied when voting outside the three central categories. The predicted probability of dissatisfaction for linear predictor η is then given by Eq. (4), with occupant-specific terms added for Model C. This quantity links the comfort models to the controllers in Part 2.

$$D(\eta) = Pr(Y \leq 2) + Pr(Y \geq 6) = \Phi(\tau_2 - \eta) + 1 - \Phi(\tau_5 - \eta) \tag{4}$$

Model fitting and comparison. Models will be fitted with the brms package in R, a Bayesian implementation of cumulative-link and multilevel models built on Stan [24]; the occupant-specific scale in Model C is implemented through brms's discrimination (disc) sub-model. A frequentist cross-check of Model B and of a random-intercept version of Model C will use the clmm function of the ordinal package [25]. Weakly informative priors will be checked with prior predictive simulation, and convergence will be assessed with R-hat (< 1.01) and effective sample sizes. Models B and C will be compared by LOOIC [26] and posterior predictive checks. Model A cannot be compared with the ordinal models by LOOIC, because its likelihood is a density over a continuous outcome while theirs is a probability over discrete categories. All three models will therefore also be compared on held-out votes using Quadratic Weighted Kappa, ordinal Mean Absolute Error and log score, after converting Model A's predictive distribution into category probabilities by integrating between category boundaries at ±0.5. Substantive differences between the models will be compared on a common scale: the implied neutral operative temperature and the temperature band within which predicted dissatisfaction stays below 10%.

Votes-per-occupant break-even analysis. Each participant's votes will be ordered in time and the final six held out as a test set. Models B and C will then be refitted using the first *n* votes of each participant, for *n* = 2, 4, 6, 8, 10, 12 and 14 (as far as the data allow), and evaluated on the held-out votes. This time-ordered design mirrors a real thermostat, which learns from past feedback and must predict future responses. Predictive performance will be plotted against *n* with uncertainty intervals; the break-even point is the smallest *n* at which Model C's improvement over Model B is credibly greater than zero. This is the key practical output of Part 1 for smart-thermostat design.

4.2 Part 2 – EnergyPlus Simulation and Controller Evaluation

Building model. A single-zone model of the field-study classroom will be built in EnergyPlus [27], using the latest stable release at project start (the version will be fixed and reported). Geometry, orientation, envelope and window properties, and HVAC type and capacity will come from the site survey and any available as-built information; the occupancy schedule will follow the actual teaching timetable, with lighting and equipment gains added. The local climate will be represented by a typical-year weather file for Hong Kong. The base case will use an ideal-loads air system to isolate the effect of the controllers; a sensitivity case will use a representative fan-coil or split-type system with part-load performance curves. A sensitivity analysis will examine the influence of uncertain envelope, infiltration, internal-gain and equipment-efficiency assumptions.

Controllers. The three controllers share a common decision rule so that differences in results reflect the comfort model rather than the control logic (Table 1). Controller 1 uses fixed heating and cooling setpoints that satisfy |PMV| ≤ 0.5 (equivalent to PPD ≤ 10%) for assumed seasonal clothing and a metabolic rate of 1.1–1.2 met. Controllers 2 and 3 choose, at each 15-minute control step during occupied hours, the most energy-favourable setpoint from a candidate grid of 20–28 °C in 0.5 °C steps — the highest cooling setpoint or the lowest heating setpoint — for which predicted dissatisfaction from Eq. (4) does not exceed 10%, matching the criterion behind Controller 1. If no candidate meets this limit, the setpoint that minimises predicted dissatisfaction is used. Controller 2 predicts dissatisfaction with Model B, treating all occupants as identical. Controller 3 averages occupant-specific predictions from Model C over the occupants present. For Controller 3, virtual classes are generated by drawing occupant-level parameters from the posterior predictive distribution of Model C, so that they represent new occupants rather than the specific participants; 50 virtual classes of 20 occupants will be simulated and the results reported as distributions across classes.

Table 1. Summary of the three candidate controllers.

Controller	Comfort model	Setpoint rule
1 – Fixed PMV	PMV/PPD (ISO 7730)	Fixed heating and cooling setpoints satisfying |PMV| ≤ 0.5 under assumed clothing and activity
2 – Population-adaptive	Model B (population ordinal)	Most energy-favourable setpoint with predicted dissatisfaction ≤ 10%, all occupants identical
3 – Personalised-adaptive	Model C (personalised ordinal)	Same rule, using the mean of occupant-specific predicted dissatisfaction over the occupants present (50 virtual classes)

Model coupling. Posterior draws of the model parameters will be exported from R as tables. A Python callback registered through the EnergyPlus Python API will read zone conditions at each timestep, evaluate Eq. (4) in closed form, and set the thermostat setpoint actuators. Because R is not called at run time, the coupling is simple, fast and reproducible, and avoids brittle file-based interchange between programs.

Performance metrics. Each controller will be simulated over a full calendar year. Reported metrics are: (i) annual HVAC energy use in kWh, disaggregated into cooling, heating and fans; and (ii) occupied-hour comfort, expressed as mean predicted percentage dissatisfied and dissatisfaction-hours (occupied hours with predicted dissatisfaction above 10%, or with |PMV| > 0.5 under the PMV criterion). Because each controller is designed to perform well under its own comfort criterion, every controller will be evaluated under all three criteria — PMV/PPD, Model B and Model C — and the results presented as a three-by-three cross-evaluation matrix. The central comparison is energy use at comparable comfort across criteria, rather than comfort under a single criterion that favours one controller by construction.

AI assistance and verification. Programming work — particularly the R statistical code and the Python controller code — will be developed with AI-assisted coding tools, with the supervisor's knowledge and consent. All model specifications, experimental design decisions and interpretations are the author's own and will be checked by the author. Verification comprises: (a) replicating the results of Favero et al. [5] (their Tables 3–7) on the same subset of the publicly available ASHRAE Global Thermal Comfort Database II as a toolchain check; (b) a parameter-recovery study, in which synthetic data with known parameters are generated for the planned design (about 20 occupants with about 20 votes each), Models B and C are refitted, and the true values are checked to lie within the estimated intervals — this also tests, before data collection, whether Model C is estimable with the planned sample; and (c) checks of the EnergyPlus model through energy-balance inspection, comparison of energy use intensity with published figures for educational buildings, and unit tests of the controller logic. A log of AI tool use will be kept for the methodology appendix of the thesis.

5. Possible Deliverables and Implementation Programme

5.1 Indicative Timeline

The project runs from mid-October 2026 to final submission and defence in about May 2027. The dates in Table 2 are indicative and may shift; the one fixed dependency is that no data, including pilot data, will be collected before ethics approval. Development of the analysis pipeline and the EnergyPlus base model is deliberately moved ahead of the field campaign, because the co-simulation work is the most likely source of delay.

Table 2. Indicative project timeline with milestones.

Period	Phase	Key tasks and milestones
Mid-Oct – Nov 2026	Foundation and ethics	Finalise revised proposal; prepare and submit HSEARS ethics application (target: early November); site survey and booking of instruments; draft questionnaire; set up R/brms toolchain; replicate Favero et al. [5]. Milestones: ethics application submitted; replication complete.
Dec 2026 – mid-Jan 2027	Pilot and pipeline development	After approval: questionnaire pilot with 2–3 classmates and one-week sensor dry run; parameter-recovery study for Models B and C; complete analysis scripts; build EnergyPlus base model with Controller 1 and the Python API coupling. Milestone: tested pipeline and base model ready before field data arrive.

Period	Phase	Key tasks and milestones
Mid-Jan – mid-Mar 2027	Classroom field campaign	Recruit participants and obtain consent; eight-week data collection; weekly data cleaning; interim model fit in Week 4 to check data quality; implement Controllers 2 and 3 with placeholder parameters. Milestone: clean dataset (Objective 1).
Mid-Mar – mid-Apr 2027	Modelling and simulation	Fit Models A–C; diagnostics; break-even curve; export posterior draws; parameterise Controllers 2 and 3; annual simulations, sensitivity analysis and cross-evaluation. Milestone: Objectives 2–5 answered.
Mid-Apr – May 2027	Writing and defence	Full thesis draft; supervisor feedback and revision; final submission and defence.

5.2 Risk Assessment and Contingency

The principal risks and the planned responses are summarised in Table 3.

Table 3. Principal risks and contingency measures.

Risk	Mitigation and contingency
Delay in ethics approval	Submit by early November, leaving about two months before the campaign. If approval is late, shorten the campaign to six weeks and reduce the break-even grid accordingly.
Low participation or too few votes per occupant	Recruit a second class if fewer than about 12 participants enrol or if the class meets only once a week. Population-level analyses (Models A and B) can be supplemented with the ASHRAE Global Thermal Comfort Database II [16]; this database is not suitable for the personalisation analysis, which requires repeated votes linked to the same participant, so a public longitudinal dataset with participant identifiers will be sought during the literature review as a fallback.
Narrow indoor temperature range	Check the range during the dry run; pair votes with the nearest station so that window-side and interior seats add spread; report the identifiability of the temperature sensitivity honestly; partial pooling shares information on the slope across occupants.
Model C does not converge or is weakly identified	Detected early by the parameter-recovery study; simplify to a random threshold shift plus random scale (dropping the random slope) and use more informative priors.
EnergyPlus Python API coupling problems	Base model and coupling built in December–January, before field data; validate on a single-zone, single-controller, short-period case before full annual runs; controllers read pre-exported parameter tables only.
Errors in AI-assisted code	Verification protocol in Section 4.2: replication of published results, parameter recovery, energy-balance checks and unit tests.

5.3 Preliminary Expected Results

The following directional hypotheses are stated before data collection and will be tested rather than assumed:

- **H1.** Models A and B will lead to different substantive estimates — in particular the implied neutral operative temperature and the temperature band with predicted dissatisfaction below 10% — and Model A will show poorer ordinal predictive performance, extending the findings of Favero et al. [5] to a classroom setting.
- **H2.** Model C will outperform Model B in predicting occupants' later votes once a modest number of votes per occupant is available, and the break-even point will lie within the range of votes collected.
- **H3.** Annual HVAC energy use in the classroom will be lower with the personalised-adaptive controller than with the fixed PMV controller at comparable predicted comfort. Whether it also outperforms the population-adaptive controller is a genuinely open question; a null result would be an acceptable and informative outcome.

6. Conclusions

This proposal describes an eight-month final-year project that quantifies how much personalised ordinal comfort models improve thermal comfort prediction and HVAC energy performance relative to conventional population-level approaches, through a coordinated classroom field study and building simulation. The project connects a recent methodological critique of thermal comfort statistics [5] with the practical design needs of smart HVAC systems, and it is scoped around measurable engineering outputs: an ethics-approved classroom dataset, three fitted comfort models, a votes-per-occupant break-even curve, a coupled EnergyPlus model and an annual comparison of three controllers.

The expected contribution is threefold. Statistically, the project extends the population-level critique of Favero et al. [5] to the individual-occupant level at which smart HVAC actually operates. Practically, it provides an estimate of how many votes per occupant are needed before personalisation pays off — a figure of direct use to smart-thermostat designers, particularly in educational buildings, where the same cohort of students occupies a classroom repeatedly. From an engineering perspective, it translates a statistical modelling choice into annual HVAC energy use and comfort metrics that building services practitioners can act on. Even if some effects prove small, the combined evidence will help identify when personalised comfort control is, and is not, worth its additional cost.

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Appendix A – Disclaimer and Originality Statement

I, LUO ZI YAN, 23073375D, hereby certify that this project proposal is entirely my own work and is original. All referenced works are properly cited and acknowledged. Where AI-assisted tools are used in the preparation of this proposal or in the execution of the project, their use is disclosed in the methodology section and a usage log is maintained, in accordance with the supervisor's and the department's guidance.

(Author's Signature) _______________LUO ZI YAN___________________

Date: _______________15/10/2026___________________

Appendix B – Summary of Revisions

This appendix lists the main changes made to the proposal submitted on 17/4/2026.

Section	**Change**	**Reason**
Whole document	Language rewritten throughout; garbled or duplicated sentences removed.	Several passages in the original were not readable.
§1, §2	Corrected attributions: the four error types are credited to Liddell and Kruschke; the scope of Favero et al.'s analysis (a single 6,048-vote dataset) is described accurately; added Schweiker et al., Wang et al., Lai and Chen and the standards cited in the text.	Accuracy of citations; the original cited "Wang et al." without a reference.
§2.5	Gap 1 narrowed to occupant-specific thresholds estimated with partially pooled models.	Group-level differences in scale use had already been reported.
§3, §4.1	Ethics: formal HSEARS application with information sheet and consent form replaces the single checkbox.	University requirement for research involving human subjects.
§4.1	Votes collected at every session (twice per session); break-even uses time-ordered hold-out and a grid of 2–14 votes.	Original design yielded at most ~16 votes per person, too few for a 20-vote grid plus test data.
§4.1	Model C reformulated as partially pooled threshold shift, scale and slope.	Six free thresholds per person are not identifiable with ~20 votes.
§4.1	LOOIC restricted to Models B vs C; Model A compared via QWK, ordinal MAE and log score.	LOOIC of a continuous and a discrete likelihood are not comparable.
§4.2	Common setpoint rule for all controllers; Controller 3 aggregation over occupants defined; three-way cross-evaluation of comfort.	Original did not define how 20 personal models give one setpoint; PMV-only evaluation favoured Controller 1.
§4.2	Verification by parameter-recovery study replaces "closed-form comparison"; replication target specified.	More informative and feasible check.
§5	Timeline rebased to Oct 2026 – May 2027; pipeline and EnergyPlus base model moved before the field campaign; campaign moved to Semester 2; risk table expanded.	Actual project dates; schedule risk.
§5.2	ASHRAE Database II retained only as a population-level fallback.	It cannot support the personalisation analysis.